Active **wildfire** detection via satellite imagery and **machine learning**: An empirical investigation of Australian wildfires

H Singh, LM Ang, SK Srivastava - Natural Hazards, 2025 - Springer

… data sources, advancing **machine learning techniques**, developing real-… **machine learning** algorithms and innovative indices like NDFI to improve **wildfire** detection and **management** …

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Wildfire induces changes in receiving waters: A **review** with considerations for water quality **management**

MJ Paul, SD LeDuc, MG Lassiter… - Water Resources …, 2022 - Wiley Online Library

… Here, we **review** the effects of **wildland fire** and destructive, urban fires in the WUI on water quality stressors important to many water quality **management** programs. While **wildfire** …

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A comparative **review** on **wildfire** simulators

GD Papadopoulos, FN Pavlidou - IEEE systems Journal, 2011 - ieeexplore.ieee.org

… This project promises to develop more reliable **bushfire** spread simulation and animation technology to underpin and support a wide range of fire **management** activities, including risk …

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Current approaches to modelling the spread of **wildland fire**: a **review**

GLW Perry - Progress in Physical Geography, 1998 - journals.sagepub.com

… Thus, this article also considers the potential application(s) of such technologies for **wildland fire management** and how such models may be integrated within land **management** …

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A **survey** on technologies for automatic **forest fire** monitoring, detection, and fighting using unmanned aerial vehicles and remote sensing **techniques**

C Yuan, Y Zhang, Z Liu - Canadian journal of forest research, 2015 - cdnsciencepub.com

… First, a brief **review** of the development and system architecture of UAV systems for **forest fire** monitoring, detection, and fighting is provided. Next, technologies related to UAV **forest fire** …

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Review and perspectives of digital twin systems for **wildland fire management**: Y. Li et al.

Y Li, T Zhang, Y Ding, R Wadhwani… - Journal of Forestry …, 2024 - Springer

… the processes of **wildfire** prevention, monitoring, disaster response, and post-fire recovery. This **review** examines the potential utility of Digital Twin in **wildfire management** and aims to …

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A **Review** on **Forest Fire** Detection **Techniques**: Past, Present, and Sustainable Future

AH Khan, AN Bahar, K Wahid - Sensors, 2026 - mdpi.com

… This paper provides a comprehensive **review** of **forest fire** detection (FFD) **techniques** and … electronic sensors, laid the foundation for modern **techniques**. Recently, there has been a …

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An overview of **machine learning** (ML) **techniques** applied to **forest fire** studies

AB Küçükarslan - Eurasian Journal of Forest Science, 2024 - dergipark.org.tr

… The prospect of using advanced ML for **forest fire management** has been widely acknowledged. Our **review** brings forth various aspects like data handling, choice of model, accuracy …

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Deep learning for **wildfire** risk prediction: Integrating remote sensing and environmental data

Z Xu, J Li, S Cheng, X Rui, Y Zhao, H He… - ISPRS Journal of …, 2025 - Elsevier

… a comprehensive **review** of **wildfire** risk prediction methodologies, with a particular focus on **deep learning** approaches combined with remote sensing. We begin by defining **wildfire** risk …

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Advancements in **forest fire** prevention: A comprehensive **survey**

F Carta, C Zidda, M Putzu, D Loru, M Anedda, D Giusto - Sensors, 2023 - mdpi.com

… prediction models is crucial for making effective decisions in **wildfire management**. There are two key strategies to achieve this goal. Firstly, it is essential to enhance model calibration …

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A **review** on **forest fire** detection **techniques**: A decadal perspective

V Chowdary, MK Gupta, R Singh - Networks, 2018 - researchgate.net

… **review** of **techniques** available for **forest fire** detection from the literature has been performed with respect to generalized parameters that fits best to all the **techniques**. … used for **forest fire** …

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A global outlook on increasing **wildfire** risk: current policy situation and future pathways

P Pandey, G Huidobro, LF Lopes, A Ganteaume… - Trees, Forests and …, 2023 - Elsevier

… forest and land **management** practices. This article aims to provide insight into **wildfire** policies, … by describing **wildfire** policies centered mainly on exclusion and **wildfire** risk mitigation. …

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Forest fire induced Natech risk assessment: A **survey** of geospatial technologies

M Naderpour, HM Rizeei, N Khakzad… - Reliability Engineering & …, 2019 - Elsevier

… literature **review** of the methods used for **forest fire** susceptibility, … and data-driven models; (b) **machine learning** models; (c) multi-… In addition, some **forest fire** detection **techniques** using …

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A performance comparison of **machine learning** models for **wildfire** occurrence risk prediction in the Brazilian Federal District region

JNS Rubí, PRL Gondim - Environment Systems and Decisions, 2024 - Springer

… This article also addresses a **review** of recent research on **wildfire** risk prediction for identifying the main features and ML models used and proposes an open dataset for the FD region. …

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An insight into **machine-learning** algorithms to model human-caused **wildfire** occurrence

M Rodrigues, J De la Riva - Environmental Modelling & Software, 2014 - Elsevier

… use of **Machine Learning** (ML) models for the assessment of human-caused **wildfire** occurrence. … for classifying **wildfire** occurrences because its calibration is extremely time-consuming. …

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Forest fire and smoke detection using **deep learning**-based learning without forgetting

VE Sathishkumar, J Cho, M Subramanian, OS Naren - Fire ecology, 2023 - Springer

… This research looks at fire/smoke detection from images using AI-based computer vision **techniques**. Convolutional Neural Networks (CNN) are a type of **Artificial Intelligence** (AI) …

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A **review** of optimization and decision models of prescribed burning for **wildfire management**

J Qi, J Zhuang - Risk Analysis, 2025 - Wiley Online Library

… Prescribed burning is an essential forest **management** tool that … In this work, we present a systematic **review** of the literature on … The **review** also examines solution **techniques**, such as …

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Social science research related to **wildfire management**: an overview of recent findings and future research needs

S McCaffrey, E Toman, M Stidham… - … Journal of **Wildland Fire**, 2013 - connectsci.au

… fire-prone communities in the Northeast: **survey** results and **wildland fire management** implications. In ‘The public and **wildland fire management**: social science findings for managers’. (…

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Local community participation and volunteerism in **wildfire** area **management**: A systematic **review**

S Kete - Turkish Journal of Forestry, 2023 - dergipark.org.tr

… and at the right cost, and this **review** deals with decision-making and planning concerning … **wildland fire management** supply chain (Martell, 2015). Transition to a new fire **management** …

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Wildfire danger prediction and understanding with **deep learning**

S Kondylatos, I Prapas, M Ronco… - Geophysical …, 2022 - Wiley Online Library

… **wildfire management**. In this work, we leverage **Deep Learning** (DL) to predict the next day's **wildfire** … of the Eastern Mediterranean and explainable **Artificial Intelligence** (xAI) to diagnose …

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